

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think>
Distance covered in the first 3 hours = $3 \times 60 = 180$ miles
Remaining distance to home = x
Time taken to cover the remaining distance = $4 - 2 - 0.5 = 1.5$ hours
Distance covered in the last 0.5 hours at 30 mph = $0.5 \times 30 = 15$ miles
Remaining distance to cover = $x - 15$
Time taken to cover this remaining distance at 80 mph = $(x - 15) / 80$
Total time taken = 2 hours (stand still traffic) + 0.5 hours (driving at 30 mph) + $(x - 15) / 80$ hours
Since the total time taken is equal to 4 hours, we can set up the equation:
 $2 + 0.5 + (x - 15) / 80 = 4$
Simplifying the equation, we get:
 $2.5 + (x - 15) / 80 = 4$
 $(x - 15) / 80 = 1.5$
 $x - 15 = 120$
 $x = 135$
So, the remaining distance to cover is 135 miles
Distance covered in the first 3 hours = 180 miles
Total distance from home = $135 + 180 = 315$ miles
</think><answer>
315 miles </answer>